

2024



AWS Enables CouponBirds to Lead in e-commerce Coupons with Excellent Data Governance Architecture

Founded in 2015, CouponBirds provides information about high-quality discounts and coupons to online shoppers worldwide. To efficiently handle the growing data traffic and improve user experience, CouponBirds adopted AWS products like Amazon Kinesis Data Streams, Amazon Data Firehose, AWS Lambda, Amazon OpenSearch Service, Amazon ELB, Amazon CloudSearch, Amazon Personalize, AWS IAM, and Amazon VPC, to build a new data governance architecture. This new data governance architecture allows CouponBirds to be better-positioned to address the needs of its users around the world while driving ongoing innovation and business growth.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Other](#)



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4 x

50%



2

improvement in data governance

cost reduction in data governance

70%

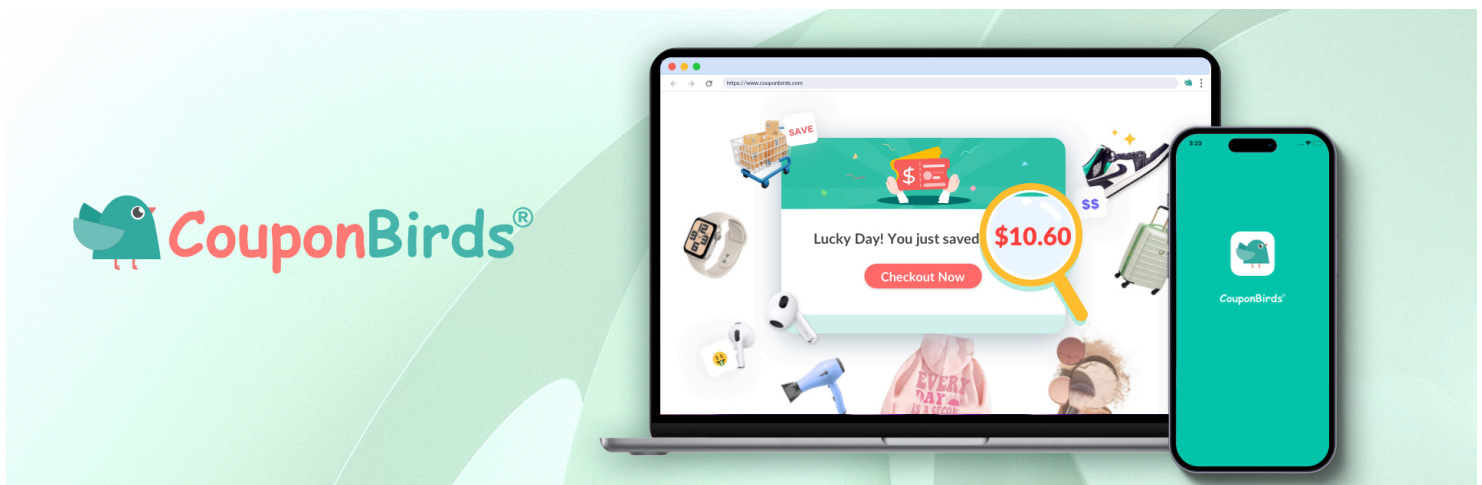
faster in responding to data changes

3 months

of a shorter product development cycle

Overview

Founded in 2015, [CouponBirds](#) provides information about high-quality discounts and coupons to online shoppers worldwide, helping millions of them get more with less each month. With a growing global user base, CouponBirds found that it was challenging to handle the heavy data traffic with its existing [data governance](#) architecture in a timely, stable and intelligent manner. Using Amazon Web Services (AWS) products and services like [Amazon Kinesis Data Streams](#), [Amazon Data Firehose](#), [AWS Lambda](#), [Amazon OpenSearch Service](#), [Amazon Elastic Load Balancing \(Amazon ELB\)](#), [Amazon CloudSearch](#), [Amazon Personalize](#), [AWS Identity and Access Management \(AWS IAM\)](#), and [Amazon Virtual Private Cloud \(Amazon VPC\)](#), CouponBirds built a new data governance architecture. The new architecture is efficient and cost-effective, taking the company's service system to a new level. It has also solved the data governance challenge arising from a growing global user base, and has been instrumental in driving innovation and growth.



Opportunity | E-commerce marketing is blustery, and technological progress is constantly affecting the development of coupon companies

In e-commerce, coupons have been a popular and effective marketing strategy. Shoppers are looking for coupons to save money, and merchants are using coupons to attract shoppers, increase sales, and improve customer loyalty. CouponBirds currently has information about the great discounts offered by 30,000-plus brands and merchants in addition to millions of users from around the world. With the continuous growth of its business, CouponBirds would need a robust and powerful infrastructure system as well as an efficient data governance architecture to take its operations and services to a new level. However, in the process of expanding and upgrading its business, the company encountered a number of technical challenges:

The difficulty and long development cycle in large-scale, real-time data governance:



CouponBirds has built a large user base and a large database of merchants thanks to 9 years in the industry. But the coupon industry is very competitive where information is changing very quickly, so it has to have real-time, efficient data governance capabilities to provide accurate and quality discount information quickly and stably for online shoppers. The company's existing server developed by itself involved a long development cycle and a growing cost. This severely hampered its development progress and business operations.

The existing services were not able to meet the new needs of users in an artificial intelligence era:

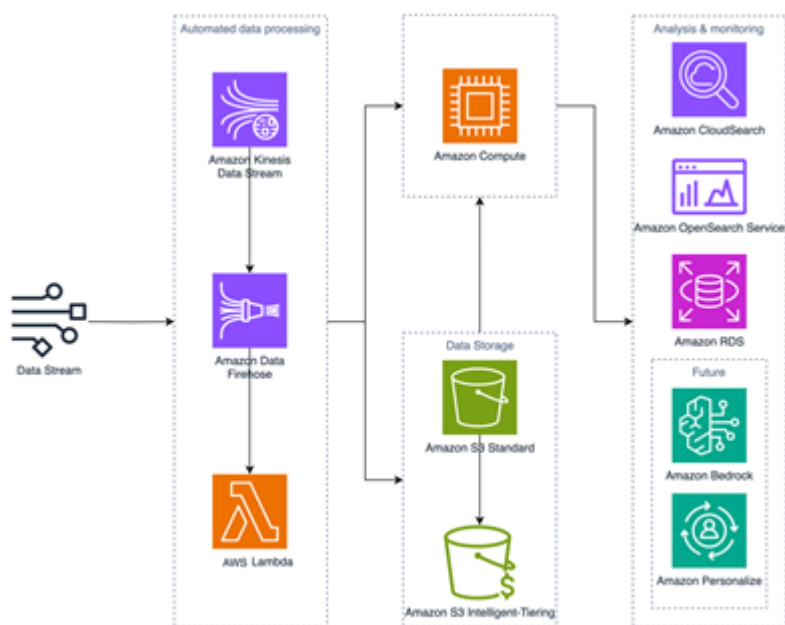
CouponBirds' challenge was not just in data governance. It was also in meeting the needs of users around the world for new services. For example, they would need more intelligent search features and more personalized recommendations. But these features would rely on cutting-edge artificial intelligence technologies, whereas developing them starting from scratch would require a huge amount of money, and it would be difficult to guarantee the outcomes too.

Low system stability/security and poor user experience during peak hours:

CouponBirds' own servers they used earlier were less stable with a high network latency as well as insufficient elasticity and flexibility, especially during peak hours like Black Friday. How to provide a smooth and stable user experience worldwide and ensure network security for users around the world was also a significant challenge.

Barry Bai, CTO of CouponBirds, said: "To address the technical challenges arising from business expansion and optimization, after thorough deliberations, we decided to upgrade our system architecture by adopting AWS products and services. Given the complete portfolio of AWS, the exceptional usability of its products, the strong synergy among them, and its professional service team, we believed that AWS could be a solid partner for us to drive business growth and technological progress."

Solution | AWS helps CouponBirds provide e-commerce coupons efficiently, stably, and innovatively



AWS helps the customer reinforce core competencies faster with its exceptional data governance solution

CouponBirds uses Amazon Kinesis Data Streams to process streaming data from various sources and uses Amazon Data Firehose to seamlessly transmit the processed data to a data warehouse and an analysis tool. This combination has effectively solved the challenge of data diversity, enhanced system compatibility and scalability, and significantly reduced the pressure of large-scale, concurrent data streams. Moreover, Amazon Kinesis Data Streams and Amazon Data Firehose are fully managed services that can be scaled up automatically, allowing CouponBirds to significantly reduce the workloads of its technical staff for operation and maintenance.

After the success in building its data governance architecture, CouponBirds continued to use the serverless computing platform AWS Lambda to process the mission-critical data with a high requirement for real-time processing, with the processed data being saved in Amazon OpenSearch Service, to meet the need for real-time data analysis and monitoring in the coupon business.

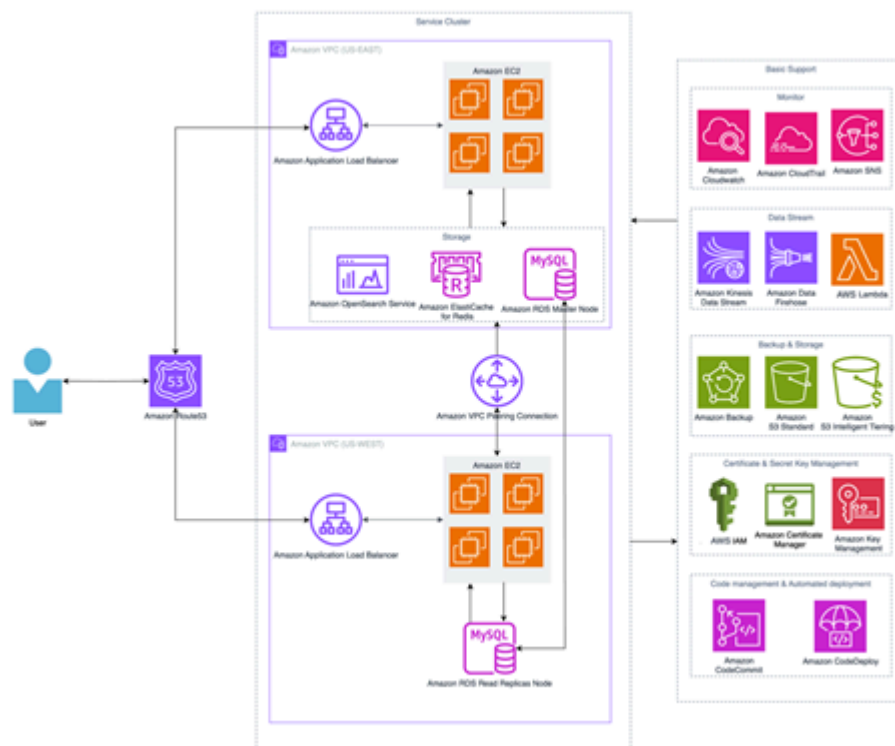
This data governance architecture is able to not only handle the pressure of processing large volumes of data for efficient and real-time data analysis, but also provide help for CouponBirds to build its competitive edge in terms of core competences.

Amazon CloudSearch and Amazon Personalize help to innovatively upgrade the user experience

To make its products and services more aligned with the needs of users today for personalized and intelligent offerings, CouponBirds adopts Amazon CloudSearch, a fully managed and customized search service that supports multiple search features such as free-text search, search in 34 languages, and highlighted display. You just need to provide a keyword, and Amazon CloudSearch will be able to automatically give you recommendations and a lot of information found. The results will be displayed for you in milliseconds, a huge enhancement of your search experience.

CouponBirds also adopts Amazon Personalize, which is based on machine learning, to make product recommendations. Thanks to these smart services, CouponBirds has been able to accelerate product development

and quickly launch some complex and innovative features, to provide a better service experience for its users around the world.



The secure global network of AWS helps the customer easily handle the traffic during peak hours

CouponBirds knows very well the importance of network stability during peak shopping periods like Black Friday and Cyber Monday. By partnering with AWS, the company quickly deployed a global network, to effectively reduce the network latency for its users around the world. CouponBirds also took advantage of Amazon ELB to flexibly handle the peak traffic, ensuring business continuity and a good user experience even when facing the challenge of high workloads with the surge of concurrent users.

Furthermore, with the significant increase of traffic during peak periods, there will be a higher security risk and a higher cost of security protection. CouponBirds improves overall system security with a lower cost by using Amazon VPC to isolate different workloads in different security zones. Additionally, it also uses the built-in services like AWS IAM to effectively reduce the money spent on purchasing, configuring, and managing various security tools. This has laid a solid foundation for expanding its global business.

Outcome | The new data governance architecture helps CouponBirds steadily expand its global business

- **A 4-time improvement of data governance and a 50% cost reduction for data processing:**

By using data stream processing tools like Amazon Kinesis Data Streams and Amazon Data Firehose, CouponBirds is able to easily handle large-scale data streams from multiple sources and improve its system's data governance by 4 times. In addition, it also uses multiple fully managed products and services of AWS to significantly reduce the cost of data governance by 50%.

- **The system is 70% faster in responding to the changing data:**

With AWS's new data governance architecture, CouponBirds is able to efficiently handle data changes in real time and address data changes 70% faster.



- Reducing product development cycle to 3 months:

The entire data governance architecture of AWS is not just easy to scale up and maintain. It also allows CouponBirds to reduce dependency on maintenance and accelerate product development. The company's product development cycle has been reduced from 8 months to 3 months.

Barry Bai, CTO of CouponBirds, said: "After using AWS products, we have substantially improved our productivity and our system's data governance, and this allows us to provide the best user experience." In the future, CouponBirds will focus on enhancing its user satisfaction and actively embrace advanced artificial intelligence technologies. The company hopes to combine the powerful artificial intelligence products of AWS with its business activities closely together to drive innovation and provide smarter services for its users around the world.

About CouponBirds

Founded in 2015, CouponBirds provides information about high-quality discounts and coupons to its users through strategic partnerships with leading brands and support for small and medium-sized enterprises . Currently, CouponBirds provides information about high-quality discounts from over 30,000 merchants, and helps millions of online shoppers save money each month.

AWS services used

Amazon Kinesis Data Streams

Amazon Kinesis Data Streams is a serverless streaming data service that makes it easy to capture, process, and store data streams at any scale.

[Learn more »](#)

Amazon Data Firehose

Amazon Data Firehose provides the easiest way to acquire, transform, and deliver data streams within seconds to data lakes, data warehouses, and analytics services.

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Amazon OpenSearch Service

Amazon OpenSearch Service makes it easy for you to perform interactive log analytics, real-time application monitoring, website search, and more.



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Amazon Personalize

Amazon Personalize accelerates your digital transformation with ML, making it easier to integrate personalized recommendations into existing websites, applications, email marketing systems, and more.

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